Ramesh Pilli

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NDSU Plant Science PhD Candidate

PROFESSIONAL SUMMARY

Applied plant breeder with Ph.D.-level training and industry experience supporting commercial breeding pipelines from early-generation population development through multi-environment evaluation, cultivar advancement, and variety release. Skilled in designing efficient field trials, making data-driven selection decisions, and integrating molecular tools to improve breeding efficiency. Proficient at translating grower, production, and agronomic performance data into practical advancement strategies. Collaborative, hands-on breeder committed to developing commercially relevant, high-performing cultivars.

CORE SKILLS

Breeding Program Development & Cultivar Advancement | Germplasm Development & Introgression | Early-Generation & Family-Based Selection | Multi-Environment Trial (MET) Design & Analysis | Marker-Assisted & Genomic Selection | Genomics | Trait Prioritization | Field Data Collection & Evaluation | Cross-Functional Collaboration (Breeding, Production, Field Operations) | Statistical Analysis (R, Python)

WORK EXPERIENCE

Doctoral Research Assistant | **North Dakota State University** ***May 2024 – Present***

- Supported cultivar advancement decisions across ~1,200 genotypes by integrating phenotypic, environmental, and genomic data to identify superior breeding lines for variety release.
- Designed and analyzed multi-environment trials (METs) using replicated and augmented designs to evaluate performance, stability, and adaptability across locations.
- Developed genomic-informed selection strategies to improve prediction accuracy and accelerate advancement cycles.
- Evaluated cold hardiness, fruit quality, and acclimation traits to guide selection of elite germplasm under production-relevant stress conditions.
- Collaborated with breeding teams to deliver clear, decision-oriented summaries supporting line advancement and germplasm prioritization.
- Conducted controlled crosses and supported population development and trait-based selection.

Camelina Breeding Intern | **Sustainable Oils, LLC** ***May 2025 – Aug 2025***

- Supported a commercial breeding pipeline from early-generation segregating populations (F1–F5) through advanced yield testing stages (R&D1, R&D2).
- Conducted single-plant and family-based selection in segregating populations to advance superior families toward variety release.
- Performed controlled crosses and introgressed disease resistance into elite germplasm.
- Collected detailed phenotypic data to support Plant Variety Protection (PVP) applications.
- Evaluated oil content using NMR spectroscopy to inform quality-focused selection decisions.
- Assisted with harvest, seed processing, and field equipment operation, gaining hands-on commercial production experience.

Graduate Research Assistant | **Washington State University** ***Aug 2023 – May 2024***

- Designed and executed field and laboratory evaluations in apple breeding populations to assess stress-related phenotypes.
- Managed large, multi-environment phenotypic datasets and supported cultivar advancement prioritization.
- Integrated genomic data to improve understanding of trait inheritance and selection efficiency.

Graduate Research Assistant | **North Dakota State University** ***May 2021 – Aug 2023***

- Designed and conducted large-scale field trials evaluating ~1,000 grape breeding lines for cold hardiness and fruit quality traits.
- Supported marker-assisted selection strategies to enhance breeding efficiency across multi-year trials.
- Constructed linkage maps and validated markers to support long-term breeding objectives.

- Assisted in population development, controlled crosses, and advancement of superior lines.

EDUCATION

Ph.D. in Plant Science (Plant Breeding & Genetics) *Expected Summer 2026*
North Dakota State University, Fargo, ND

Research Experience in Genomics *Aug 2024 – May 2025*
Washington State University, Pullman, WA

M.S. in Plant Science (Plant Breeding & Genetics) *2023*
North Dakota State University, Fargo, ND

B.S. in Agriculture *2019*
Acharya N.G. Ranga Agricultural University, India

TECHNICAL SKILLS

Programming & Analytics: R, Python (pandas, matplotlib), Excel
Genomics & Molecular Tools: GWAS, Genomic Selection, Marker-Assisted Selection (MAS), RNA-seq, Variant Calling
Visualization & Reporting: ggplot2, Power BI (breeding & yield dashboards)
Systems & GIS: Linux, Bash, HPC, QGIS, ArcGIS

PEER-REVIEWED PUBLICATIONS

Kadium, V.R., **Pilli, R.**, Svyantek, A., Wang, Z., Stenger, J., Bhoomireddy, R., Auwarter, C., Li, X., & Hatterman-Valenti, H. (2026). Dissecting the Genetic Basis of Relevant Fruit Quality Traits in Interspecific Grapevines (*Vitis* spp.). *Horticulture Research*, uhaf353. https://doi.org/10.1093/hr/uhaf353

Svyantek, A., Stenger, J., Auwarter, C., Shikanai, A., Kose, B., Wang, Z., Kadium, V.R., Brooke, M., Delavar, H., **Pilli, R.**, Bogenrief, S., & Hatterman-Valenti, H. (2024). This is how we chill from '23 'til: breeding cold-hardy grapevines for unprecedented and unpredictable climate challenges. *ISHS Acta Horticulturae*, 1385. https://doi.org/10.17660/ActaHortic.2024.1385.17

CERTIFICATIONS

- **Precision Agriculture & GIS Mapping** – Coursera
- **Big Data Applied Statistical Analysis Certificate** – North Dakota State University (2024)
  Coursework: Applied Statistical Machine Learning (STAT 712) · Basic Computational Statistics using R (STAT 711) · Introduction to Data Science (STAT 713) · Statistical Big Data Visualization (STAT 714)

SELECTED PRESENTATIONS & ACHIEVEMENTS

- Presented applied genomic and phenotypic analyses identifying genetic regions associated with sunburn susceptibility in apple breeding populations — Plant & Animal Genome Conference (PAG31), 2024.
- Delivered GWAS results identifying loci associated with cold hardiness in interspecific wine grape breeding populations.
- Reported genomic analyses of basic fruit chemistry traits in cold-climate wine grape breeding populations.

AWARDS & SCHOLARSHIPS

- Plant and Animal Genome Conference (PAG31) Travel Grant, 2024
- American Society of Enology and Viticulture – Graduate Scholarship, 2022–2025
- Charles and Linda Moses Presidential Graduate Fellowship, North Dakota State University, 2022–2024